

Rachel Wegener

 [rwegener2](#)
 [/rachel-wegener](#)

 Vancouver, BC
 US Citizen

About

Geospatial software engineer with 5+ years of experience making data accessible for people.

Work Experience

Cloud Infrastructure & Coding Education Lead

Jan. 2024 – Sep. 2025

NASA Student Airborne Research Program (SARP)

Washington, DC

- > Led the implementation of a JupyterHub cloud platform to support 48 earth science research projects
- > Managed two coding instructors
- > Led programming curriculum development for Python and Open Science

Tools: Python, Docker, conda, JupyterLab, JupyterBook, git

Open Source Python Developer

Jun. 2023 – Sep. 2023 (part time)

University of New Hampshire

Washington, DC (remote)

- > Furthered development of the icepyx Python library, an open source library for accessing large-scale data from NASA's ICESat-2 lidar satellite
- > Collaboratively designed and implemented a module for enabling cloud data access of NASA's ICESat-2 data. Refactored authentication module.

Tools: Python, pytest, s3fs, spatial libraries (ex. shapely, geopandas), HDF and netCDF tools

Cloud Engineer; Project Manager

Jan. 2020 – May 2021

Development Seed

Denver, CO (remote)

- > Researched, selected, and implemented the most applicable technologies to process and archive data based on the needs of each project, with a heavy emphasis on open source tools and technologies.
- > Built cloud systems for data processing in AWS
- > Worked with project partners on applied science projects to develop shared goals that satisfy diverse needs and constraints
- > Confront NASA organization-level challenges such as metadata and cataloging, diverse scientific use cases, cloud costs, and strengths/weaknesses of emerging cloud file formats
- > Participated in collective ownership of the company by identifying and pursuing initiatives that better the organization
- > Managed project teams to ensure that developers had guidance and support. Ensured project deadlines were communicated to partners and met.
- > Assisted in business development opportunities and project proposals for new work

Tools: Python, FastAPI, pydantic, sqlalchemy, AWS Cloud Development Kit, STAC specification

GeoPython Developer, Emerging Technologies

Jul. 2018 – Dec. 2019

Maxar

Westminster, CO

- > Worked on a team of 6 to prototype new geospatial analytics products on top of DigitalGlobe high resolution satellite imagery. Integrated external public data sources
- > Built scaled data processing solutions
- > Developed algorithms for new capabilities using remote sensing and data science technologies

Tools: Postgres Databases, AWS Cloud, Python, psycopg2, luigi, spatial libraries (ex. shapely, pyproj), GDAL

Education

University of Maryland, College Park

2021 – 2023

M.S. Atmospheric & Oceanic Science

College Park, MD

University of Denver

2012 – 2017

B.S. Physics; Environmental Science

Denver, CO

Selected Open Source Projects

Editor, Journal of Open Source Software

Jan. 2024 – present

- Oversee submissions to the Earth Science & Ecology track. Locate appropriate editors, spot check software quality, and facilitate an open and collaborative review process.

Open source developer, icepyx

Jun. 2023 – present

- Build and maintain the cloud data read module
- Contribute as needed to other cloud-related tools in the ecosystem, including earthaccess and xarray

Participant, Climate Community Hackweeks

- Worked with community members of all skill levels and backgrounds to prototype new tools and reach consensus on community needs
- Ocean Hackweek (2021) – ICESat-2 Hackweek (2023) – ICESat-2 Hackweek *developer only, by invitation* (2025)

Open source contributor, Pangeo Forge

Jun. 2021 – Jun. 2022

- Wrote documentation and tutorials for the newly developed Pangeo Forge software, a scientist-friendly ETL engine for converting data into cloud-optimized file formats

Dataset contributor, AWS Public Datasets (NASA SpaceApps)

Jun. 2020

- Converted 15+ years of data across 6 NASA products to Cloud Optimized GeoTIFF

Skills

Programming Python, Docker, Amazon Web Services, git/Github, postgres/postGIS, Bash/shell, Linux, QGIS

Relevant Coursework (U of Maryland) Intro to Computer Systems, Intro to Parallel Computing

Communities & Education CSCCE Open Source Governance Certificate, U of Maryland Teaching and Learning Center Active Learning Group, NASA TOPS Open Science Certification

Languages English (native), Spanish (fluent), French (intermediate), Arabic (intermediate)

Selected Publications

Scheick, Jessica, Anthony Arendt, Andrew Barrett, Zachary Fair, Scott Henderson, Miguel Jimenez, Zachary Katz, Joseph Kennedy, Luis Lopez, Owen Littlejohns, Chance Roberts, Benjamin Smith, Tasha Snow, JP Swinski, and Rachel Wegener. “Ten Simple Rules to Increase the Usability of Earth Observation Datasets”. In: (*in prep*) (2026).

Wegener, Rachel, Jacob Wenegrat, Veronica P. Lance, and Skylar Lama. “Spatial Variability of Marine Heatwaves in the Chesapeake Bay”. In: *Estuaries and Coasts* 48.113 (2025). ISSN: 1559-2731. DOI: [10.1007/s12237-025-01546-9](https://doi.org/10.1007/s12237-025-01546-9).

Stern, Charles, Ryan Abernathey, Joseph Hamman, Rachel Wegener, Chiara Lepore, Sean Harkins, and Alexander Merose. “Pangeo Forge: Crowdsourcing Analysis-Ready, Cloud Optimized Data Production”. In: *Frontiers in Climate* 3 (2022). ISSN: 2624-9553. DOI: [10.3389/fclim.2021.782909](https://doi.org/10.3389/fclim.2021.782909).

Selected Presentations

Wegener, Rachel, Disha Sardana, and Riley McCue. “Filling a Skills Gap: Teaching Programming to Earth Scientists”. In: Washington, DC: American Geophysical Union, 2024.

Wegener, Rachel and Jessica Scheick. “Using Icesat-2 for Data Access”. In: online: The International Colloquium on Space and Sustainability, 2023.

Wegener, Rachel and Charles Stern. “Pangeo Forge: An ETL Pipeline for Cloud-Optimized Analysis Ready Data”. In: online: AGU Ocean Sciences Meeting, 2022.

Wegener, Rachel. “STAC and Cloud-Optimized Data: Publicly Accessible COGS for Web Data Exploration”. In: online: Earth Systems Information Partners (ESIP) Summer Meeting, 2020.